

Project Execution Files

DATE	29 September 2026
TEAM ID	SWTID-2026-1812
PROJECT NAME	Comic craft AI comic story creator using Gemini model
MAXIMUM MARK	3 Mark

Project Configuration

The ComicCraft – AI Comic Story Creator project is configured as a web-based application using Python, FastAPI, HTML, CSS, Jinja2 templates, Google Gemini API, and image generation services.

Environment Configuration

Configuration	Details
Operating System	Windows
Programming Language	Python
Backend Framework	FastAPI
Frontend	HTML, CSS, Jinja2
AI Service	Google Gemini API
Image Generation	AI-based image generation service
PDF Generation	Python PDF generation library
Development Server	Uvicorn
Browser	Google Chrome / Microsoft Edge
Project Directory	ComicCraftAI

Required Project Files

The following files and folders are required for running the project:

```
comiccraft/
|
├─ app/
|   ├── main.py
|   ├── routes.py
|   ├── gemini_flash.py
|   ├── gemini_pro.py
|   ├── image_generator.py
|   ├── layout_builder.py
|   └── exporters.py
|
├─ templates/
|   ├── index.html
|   ├── comic_preview.html
|   └── export_success.html
|
├─ static/
|   ├── panels/      # Generated panel images
|   ├── exports/     # Exported comic PDFs
|   └── fonts/       # Fonts for PDF generation
|
├─ .env              # Environment variables (API keys)
└─ requirements.txt  # Python dependencies list
```

Configuration Instructions

1. Install Python on the local computer.

2. Open the ComicCraftAI project folder.
3. Create a Python virtual environment if required.
4. Install the dependencies listed in `requirements.txt`.
5. Create a `.env` file in the project root folder.
6. Add the required API credentials to the `.env` file.
7. Do not share or upload the actual API secret key to GitHub.
8. Check that the required project folders are available.
9. Start the FastAPI application using Uvicorn.

API Configuration

The application requires an API key for AI-based content generation. The API key is stored securely in the `.env` configuration file and is accessed by the application through environment variables.

Important: The actual API secret key must not be included in the project documentation, screenshots, GitHub repository, or public submission.

Project Execution

Step 1 – Start the Application

Open the project directory in Command Prompt or Visual Studio Code and run:

```
python -m uvicorn app.main:app --reload
```

Step 2 – Open the Application

After the server starts successfully, open the local application URL in a web browser:

```
http://127.0.0.1:8000
```

Step 3 – Enter Project Input

Enter the required story information, such as:

- Story prompt
- Main character
- Setting

- Tone
- Art style

Step 4 – Generate Comic

Click the **Generate Comic** button. The application processes the input and generates the story outline, narration, dialogue, and comic-panel images.

Step 5 – Preview and Export

The generated comic is displayed on the preview page. The user can review the panels and use the PDF export/download option to obtain the final comic.

Project Execution

Submission Checklist

S.No	Essential Item	Status
1	Project Source Code	Completed
2	Project Configuration / Setup Files	Completed
3	Requirements File (requirements.txt)	Included
4	Environment Configuration (.env)	Configured
5	README File (README.md)	Included

6	Project Folder Structure	Completed
7	Project Execution Instructions	Completed
8	Functional Testing	Completed
9	Performance Testing	Completed
10	Testing Screenshots / Evidence	Included
11	Final Project Output	Completed
12	Generated Comic PDF	Included
13	Demonstration Video	Ready
14	GitHub Repository	Updated
15	Project Documentation	Completed
16	Final Submission Verification	Completed

7.3 Deployment / Application URL

Local Application URL

http://127.0.0.1:8000

The above URL is used for local development and demonstration. If the application is deployed to a cloud hosting service, the final deployment URL can be added here.

7.4 Troubleshooting and Support

Issue	Possible Solution
Application does not start	Check Python installation and project dependencies.
API error	Check the API configuration in the <code>.env</code> file.
Page does not load	Check whether the Uvicorn server is running.
Images are not generated	Check the image-generation configuration and internet connection.
PDF is not generated	Check the export folder and PDF-generation dependencies.
Template error	Check the HTML/Jinja2 template files and their locations.

Resources

- Project source code
- `requirements.txt`
- `.env` configuration file
- Project README
- Test screenshots
- Generated comic PDF
- Project demonstration video